Worksheet 4.3 — Critical, Inflection & End Behaviour

Grade 12 Calculus and Vectors (MCV4U) · Unit 4: Curve Sketching

The Big Ideas

Gather all features before sketching: domain, intercepts, asymptotes, extrema (from f'), inflection/concavity (from f''), and end behaviour.

- From f': increase/decrease and extrema.
- From f'': concavity and inflection.
- From f: domain, intercepts, asymptotes, end behaviour.

The Big Picture

Gathering the features

A full analysis collects critical numbers, inflection points, intercepts, and the behaviour as $x \to \pm\infty$.

Asymptotes and limits

Limits at infinity reveal horizontal asymptotes, while limits at breaks reveal vertical ones.

A complete picture

Together these features describe every important part of the graph before you draw it.

Worked Examples

Example 1 — List features

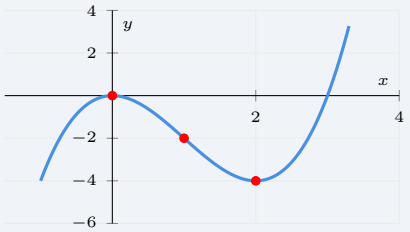
Key features of $f(x) = x^3 - 3x^2$.

Solution. **Step 1 (find extrema from f'):** $f'(x) = 3x^2 - 6x = 3x(x-2) = 0 \Rightarrow x = 0, 2$; testing signs gives max $(0, 0)$ and min $(2, -4)$.

Step 2 (find inflection from f''): $f''(x) = 6x - 6 = 0 \Rightarrow x = 1$, giving $f(1) = -2$, so $(1, -2)$.

Step 3 (end behaviour): odd-degree leading term x^3 with positive coefficient $\Rightarrow$ down on the left, up on the right.

Conclusion: max $(0, 0)$, min $(2, -4)$, inflection $(1, -2)$, ends down–up:



Example 2 — Read f from f'

$f' > 0$ on $(-\infty, 1)$, $f' < 0$ on $(1, \infty)$. Describe f near $x = 1$.

Solution. **Step 1 (interpret the sign pattern):** f' goes from positive to negative at $x = 1$.

Step 2 (apply the first-derivative test): a $+ \to -$ change means f switches from increasing to decreasing.

Conclusion: local max at $x = 1$.

Example 3 — Classify from f"

$f'(2) = 0$ and $f''(2) = -3$. Classify $x = 2$.

Solution. **Step 1 (confirm it's a critical point):** $f'(2) = 0$.

Step 2 (apply the second-derivative test): $f''(2) = -3 < 0$, so the curve is concave down there.

Conclusion: local maximum at $x = 2$.

Example 4 — Rational asymptotes

Asymptotes of $f(x) = \dfrac{x}{x-2}$.

Solution. **Step 1 (find the vertical asymptote, where the denominator is 0):** $x-2 = 0 \Rightarrow x = 2$.

Step 2 (find the horizontal asymptote by comparing degrees): numerator and denominator both have degree 1, so the HA is the ratio of leading coefficients, $\frac{1}{1} = 1$.

Conclusion: vertical asymptote $x = 2$; horizontal asymptote $y = 1$.

Example 5 — Inflection, not extremum

Inflection of $f(x) = x^3 + 1$; why no extremum?

Solution. **Step 1 (find f'' and set to 0):** $f''(x) = 6x = 0 \Rightarrow x = 0$, giving $f(0) = 1$, so $(0, 1)$.

Step 2 (check f' to see if it's also an extremum): $f'(x) = 3x^2 \geq 0$ for every x — never negative, so no sign change.

Conclusion: $(0, 1)$ is an inflection point but not an extremum, since f' never changes sign there.

Example 6 — Critical numbers

Critical numbers of $f(x) = x^3 - 3x^2$.

Solution. **Step 1 (differentiate):** $f'(x) = 3x^2 - 6x$.

Step 2 (factor and set to 0): $3x(x-2) = 0$.

Conclusion: $x = 0$ and $x = 2$.

Example 7 — Classify from f"

$f'(3) = 0$, $f''(3) > 0$. Classify $x = 3$.

Solution. **Step 1 (confirm it's a critical point):** $f'(3) = 0$.

Step 2 (apply the second-derivative test): $f''(3) > 0$ means concave up there.

Conclusion: local minimum at $x = 3$.

Example 8 — Inflection

Inflection of $f(x) = x^3 - 3x^2$.

Solution. **Step 1 (find f''):** $f''(x) = 6x - 6$.

Step 2 (set $f'' = 0$ and solve): $x = 1$, giving $f(1) = 1 - 3 = -2$.

Conclusion: inflection point at $(1, -2)$.

Example 9 — End behaviour

End behaviour of $f(x) = -x^3 + 2x$.

Solution. **Step 1 (identify the leading term):** the highest-degree term is $-x^3$, degree 3 (odd) with negative coefficient.

Step 2 (recall the pattern for odd-degree, negative leading coefficient): the graph rises on the far left and falls on the far right.

Conclusion: up–down end behaviour.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Critical numbers of $f(x) = x^3 - 3x^2$?

Question 2

If $f'(3) = 0$ and $f''(3) > 0$, classify $x = 3$.

Question 3

Inflection point of $f(x) = x^3 - 3x^2$?

Question 4

Asymptotes of $f(x) = \dfrac{1}{x-3}$?

Question 5

End behaviour of $f(x) = -x^3 + 2x$?

Question 6

Local extrema of $f(x) = x^3 - 3x$?

Question 7

Asymptotes of $f(x) = \dfrac{x}{x-2}$?

Question 8

If $f'(1) = 0$ and $f''(1) < 0$, classify $x = 1$.

Question 9

Inflection point of $f(x) = x^3$?

Question 10

End behaviour of $f(x) = x^4 - 1$?

Question 11

Where is $f(x) = x^3 - 3x^2$ increasing?

Question 12

Does $f(x) = x^3 + 1$ have a local extremum?

Question 13 — Challenge

List every key feature of $f(x) = x^3 - 3x$ (intercepts, extrema, inflection, ends).

Question Recap & Answer Key

Question Recap

1. Critical numbers of $f(x) = x^3 - 3x^2$?
2. If $f'(3) = 0$ and $f''(3) > 0$, classify $x = 3$.
3. Inflection point of $f(x) = x^3 - 3x^2$?
4. Asymptotes of $f(x) = \dfrac{1}{x-3}$?
5. End behaviour of $f(x) = -x^3 + 2x$?
6. Local extrema of $f(x) = x^3 - 3x$?
7. Asymptotes of $f(x) = \dfrac{x}{x-2}$?
8. If $f'(1) = 0$ and $f''(1) < 0$, classify $x = 1$.
9. Inflection point of $f(x) = x^3$?
10. End behaviour of $f(x) = x^4 - 1$?
11. Where is $f(x) = x^3 - 3x^2$ increasing?
12. Does $f(x) = x^3 + 1$ have a local extremum?
13. List every key feature of $f(x) = x^3 - 3x$ (intercepts, extrema, inflection, ends).

✓ Answer Key

1. $x = 0, 2$
2. local min
3. $(1, -2)$
4. VA $x = 3$, HA $y = 0$
5. updown
6. max $(-1, 2)$, min $(1, -2)$
7. VA $x = 2$, HA $y = 1$
8. local max
9. $(0, 0)$
10. upup
11. $x < 0$ or $x > 2$
12. no (inflection at $(0, 1)$)
13. x-int $0, \pm\sqrt{3}$; max $(-1, 2)$; min $(1, -2)$; inflection $(0, 0)$; ends downup